



ETHICS AND ECONOMICS OF INSTITUTIONAL GOVERNANCE

Winter Term 2024/25, 2nd Trial

- Please answer **all** questions. Maximum: **60 points**, time to answer the questions: **60 minutes**
- Permitted tools: **non**. In the case of cheating or deception, the exam will be graded as failed.
- Please answer in complete sentences, not bullet points. Indicate the problem number with your answers.

Problem 1: Lecture 10 (literature, 30 points)

William Nordhaus (2019) focuses on climate change in his research. Please, describe his main arguments by answering the following sub-questions:

- a) What is a global externality, and why is climate change considered the most significant of such externalities? Please explain. (5 points)
- b) Describe the role of Integrated Assessment Models (IAMs) in addressing climate change, using the DICE model as an example. (5 points)
- c) What are the three primary strategies to address climate change mentioned in the text? (5 points)
- d) William Nordhaus criticizes the political practice of setting an arbitrary temperature target for global climate policy. What is his main argument? (5 points)
- e) According to William Nordhaus, the core problem of addressing climate change can be described as a global public good. What are the main implications of such a description? (5 points)
- f) Why does William Nordhaus propose to mitigate the climate change problem with the help of a universal carbon price that should increase in the course of time? Please explain (5 points)



Problem 2: Karl Marx and the labor market (lecture 12, 30 points)

Karl Marx and Friedrich Engels argue in the Communist Manifesto that workers are exploited. The two authors apply findings valid in the goods market to the labor market. The neoclassical analysis disagrees with the conclusion of the two authors. Please reconstruct the argumentation by answering the following sub-questions.

- a) Marx and Engels argue that because of competition between workers, the wage is always equal to the subsistence minimum. A worker who demands payment above the subsistence wage is displaced by an unemployed worker. As a result, companies are expropriating the value added as unpaid work. Please reconstruct Marx and Engels' argument in an appropriate labor market diagram. What is the wage for the workers? Suppose there is an increase in demand. Who benefits from this increase and why? (12 points)

- b) The neoclassical analysis shows that Marx and Engels made an important mistake in their analysis. They do not take into account that work is exhaustible. Thus, Marx and Engels do not see that work becomes a scarce factor in capitalism. Companies compete for workers. As a result, companies have to pay a wage that provides a large part of the added value of the production to the workers. Please reconstruct this situation as a social dilemma. Explain the strategies of the actors in this dilemma. Determine the ordinal payoffs as well as the outcome of this game. Is the outcome Pareto-superior or Pareto-inferior for the involved actors? (12 points)

- c) Still today, the argument that workers are exploited dominates the public debate. As a solution, Germany introduced the minimum wage a few years ago. Many economists are skeptical about the minimum wage. Please briefly outline the economic objections to a minimum wage. Given these objections, can you think of a superior alternative to help workers? Please explain. (6 points)